Composition

Composite function: for $G : \mathbb{R}^n \to \mathbb{R}^p$ and $F : \mathbb{R}^p \to \mathbb{R}^m$,

$$(F \circ G)(\mathbf{x}) = F(G(\mathbf{x})), \qquad F \circ G : \mathbb{R}^n \to \mathbb{R}^m.$$

Apply G first, then F. With restricted domains, require $\mathbf{x} \in \mathrm{dom}(G)$ and $G(\mathbf{x}) \in \mathrm{dom}(F)$. Order matters: $F \circ G$ and $G \circ F$ need not agree or both be defined.

One independent variable

Scalar chain rule: $(f \circ g)'(t_0) = f'(g(t_0))g'(t_0)$, when g is differentiable at t_0 and f is differentiable at $g(t_0)$.

Scalar function along a curve: let $\mathbf{r} : \mathbb{R} \to \mathbb{R}^2$ be differentiable at t_0, let $\mathbf{a} = \mathbf{r}(t_0)$, and let $f : \mathbb{R}^2 \to \mathbb{R}$ be differentiable at $\mathbf{a}$. Then

$$(f \circ \mathbf{r})'(t_0) = f_x(\mathbf{a})r_1'(t_0) + f_y(\mathbf{a})r_2'(t_0) = \nabla f(\mathbf{a}) \cdot \mathbf{r}'(t_0).$$

Differential notation: for $z = f(x(t), y(t))$, $\dfrac{dz}{dt} = \dfrac{\partial f}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial f}{\partial y}\dfrac{dy}{dt}$; evaluate the partials at $(x(t), y(t))$.

Two independent variables

Let $G : \mathbb{R}^2 \to \mathbb{R}^2$ be differentiable at $\mathbf{a}$, and $f : \mathbb{R}^2 \to \mathbb{R}$ differentiable at $\mathbf{b} = G(\mathbf{a})$.

$$(f \circ G)_x(\mathbf{a}) = f_x(\mathbf{b})(G_1)_x(\mathbf{a}) + f_y(\mathbf{b})(G_2)_x(\mathbf{a}),$$
$$(f \circ G)_y(\mathbf{a}) = f_x(\mathbf{b})(G_1)_y(\mathbf{a}) + f_y(\mathbf{b})(G_2)_y(\mathbf{a}).$$

Generalized chain rule

Theorem: let $G : \mathbb{R}^n \to \mathbb{R}^p$ be differentiable at $\mathbf{a}$, and $F : \mathbb{R}^p \to \mathbb{R}^m$ differentiable at $G(\mathbf{a})$. Then $F \circ G$ is differentiable at $\mathbf{a}$, and

$$J_{F\circ G}(\mathbf{a}) = J_F(G(\mathbf{a}))J_G(\mathbf{a}).$$

Matrix sizes: $J_F(G(\mathbf{a}))$ is $m \times p$ and $J_G(\mathbf{a})$ is $p \times n$; their product is $m \times n$.

Component form: for $1 \le i \le m$, $1 \le j \le n$, with $y_1, \ldots, y_p$ the input coordinates of F,

$$\frac{\partial (F_i \circ G)}{\partial x_j}(\mathbf{a}) = \sum_{k=1}^{p} \frac{\partial F_i}{\partial y_k}(G(\mathbf{a}))\frac{\partial G_k}{\partial x_j}(\mathbf{a}).$$

Order and evaluation: the rightmost matrix acts first. Evaluate the inner Jacobian at $\mathbf{a}$ and the outer one at $G(\mathbf{a})$.

Column-gradient form: for scalar-valued $f : \mathbb{R}^p \to \mathbb{R}$ under the same differentiability hypotheses,

$$\nabla(f \circ G)(\mathbf{a}) = J_G(\mathbf{a})^T \nabla f(G(\mathbf{a})).$$

The transpose is required because gradients are column vectors; $J_f = (\nabla f)^T$.